

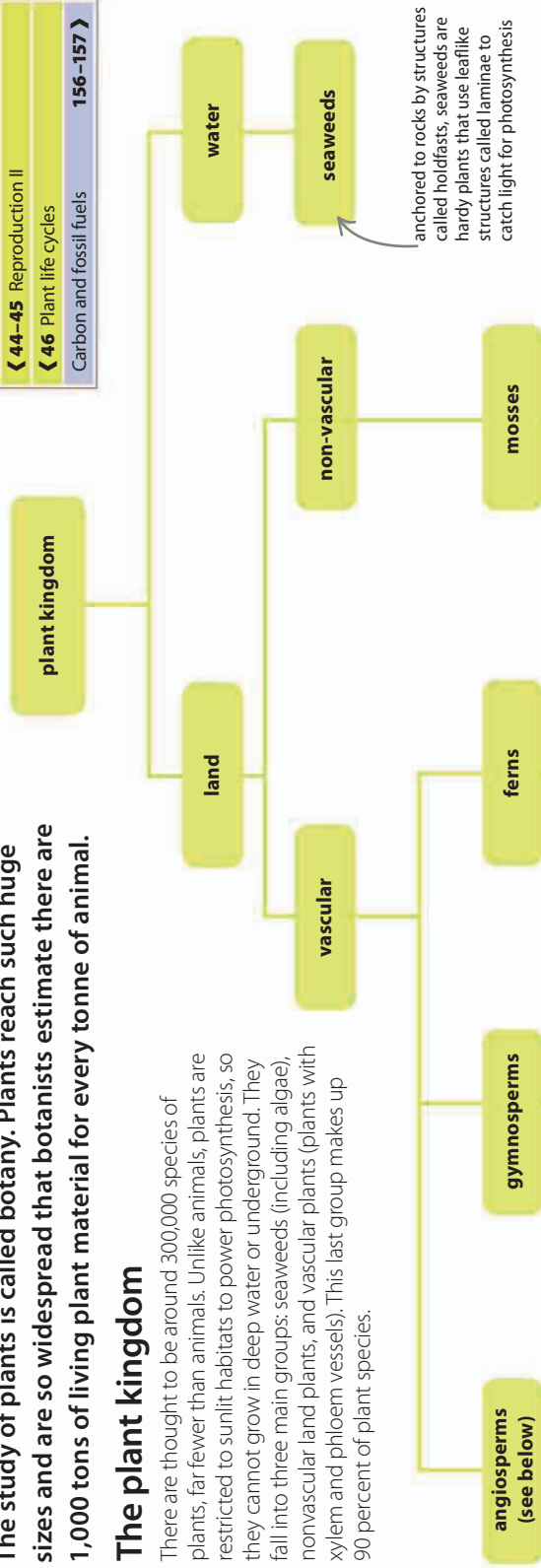
Plants

THE ORGANISMS THAT MAKE UP THE PLANT KINGDOM RANGE FROM SIMPLE MOSSES TO COMPLEX FLOWERING PLANTS.

The study of plants is called botany. Plants reach such huge sizes and are so widespread that botanists estimate there are 1,000 tons of living plant material for every tonne of animal.

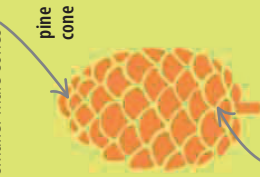
The plant kingdom

There are thought to be around 300,000 species of plants, far fewer than animals. Unlike animals, plants are restricted to sunlit habitats to power photosynthesis, so they cannot grow in deep water or underground. They fall into three main groups: seaweeds (including algae), nonvascular land plants, and vascular plants (plants with xylem and phloem vessels). This last group makes up 90 percent of plant species.



angiosperms (see below)

opening in top of female cone receives pollen from smaller male cones

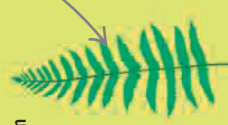


pollen develops into a seed inside the cone, before the cone lets the seed fall to the ground

◁ Gymnosperms

The first plants to reproduce using seeds were the gymnosperms, which include today's conifer trees, such as pines, cycads, and firs. Their scientific name means "naked seed," which refers to the way the seeds are not enclosed in a coat or fruit, as in the flowering plants (angiosperms) that evolved later.

structures on the fronds called sporangia disperse spores—which form new ferns

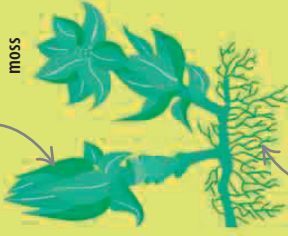


a young bud unfurls into a frond

◁ Ferns

Ferns are primitive vascular plants that do not produce seeds. They have roots and stems with bundles of xylem and phloem. These not only transport water and sugars, but also provide enough support for the plants to reach large sizes. Ferns include the first "trees", about 350 million years ago.

simple stems are flattened to catch light



the plant is held in place by extensions called rhizoids

◁ Mosses and liverworts

The most simple land plants are the mosses and liverworts. They do not have true leaves or roots. Without xylem or phloem to transport material, they are restricted in size and require a damp habitat. Mosses and liverworts reproduce using eggs and sperm that swim between plants.

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